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Unsupervised Representation Learning of Acoustic Emission for Battery Diagnostics with VQ-VAE

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Abstract

Acoustic emission (AE) provides continuous, noninvasive monitoring of micromechanical events during battery cycling. Prior analyses of battery AE rely on handcrafted signal features and heuristics, yet AE waveforms are mixtures of overlapping sources and propagation paths, and labeled datasets are scarce. These factors motivate unsupervised representation learning that can operate directly on raw waveforms.

We study a 1D Vector-Quantized Variational Autoencoder (VQ-VAE) trained on AE events from two full lithium-ion experiments. The model is trained on a 2 mA reference run and evaluated on a 5 mA run to probe generalization. The VQ-VAE achieves stable reconstruction (median MSE 8.1×10^{-7} and NMSE 0.0034 on the 2 mA set; 4.1×10^{-7} and 0.0206 on the 5 mA set) and learns diverse codebook usage (perplexity $\approx$40.9 vs. $\approx$51.7, with distinct high-frequency codes across runs). These discrete codes enable bag-of-codes analysis and downstream clustering without labels.

Our results indicate that VQ-VAE gives compact, label-free representations that are sensitive to experimental conditions while remaining reconstructive. This establishes a practical baseline for unsupervised AE analytics and sets up physics-informed clustering to relate latent patterns to electrochemical behavior.

1 Background

Acoustic emission (AE) records short elastic bursts from micro-mechanical events inside a cell. It works in situ and in real time, which makes AE useful for diagnosis and monitoring.

AE has a long history in structural health monitoring (bridges, concrete, and rock) [1]. In batteries, interpretation is harder: signals are weak, sparse, and noisy [2]. Waveforms mix multiple sources and propagation paths. Labels are scarce. Reviews summarize sensor types, coupling, typical preprocessing, and reported links between AE activity and electrochemical phases or safety-relevant events [3].

Recent studies explore supervised and unsupervised ML on AE to detect anomalies, cluster patterns, and relate activity to cycling or abuse tests. For cylindrical cells, AE+ML has been used to separate operating regimes and flag outliers under fast charge or abuse [2]. The consensus is clear: ML improves scalability and sensitivity, yet noise, domain shift, and limited labels remain central challenges [3].

End-to-end representation learning addresses these issues by feeding raw waveforms to a model and allowing it to learn task-useful features, rather than relying on handcrafted ones. In our setting, the pipeline is waveform $\rightarrow$ VQ-VAE $\rightarrow$ discrete codes / bag-of-codes $\rightarrow$ clustering or electrochemical alignment. Vector-quantized VAEs introduce a discrete bottleneck: a codebook maps continuous latents to indices [4]. Because these representations emphasize distributions rather than point assignments, the resulting count-based summaries (code indices and bag-of-codes) capture the near-continuous geometry observed in AE data better than hard clusters. They also transfer across experiments and compose naturally with simple statistics.

This study applies a 1-D VQ-VAE to lithium-ion AE data. Each event is a 3000-sample waveform around the trigger. We train on one experiment and evaluate on another to probe generalization. We report reconstruction error and codebook usage (perplexity). We also analyze how discrete codes vary with operating condition and over cycling time, and outline alignment with electrochemical features (e.g., voltage windows, dQ/dV).

Contributions

- An end-to-end pipeline for unsupervised AE representation: raw waveform $\rightarrow$ VQ-VAE discrete codes $\rightarrow$ bag-of-codes and clustering.
- Cross-experiment evaluation showing stable reconstructions and distinct code usage patterns, indicating sensitivity to operating conditions.
- A practical analysis recipe to link discrete codes to electrochemical context, enabling physics-informed clustering and weak supervision.

2 Methodology

2.1 Data

2.1.1 Cell configuration and cycling protocol

We analyze coin-type half cells with a $LiFePO_4$ (LFP) vs. TiS_2 electrode pairing. Assembly metadata in the file headers indicate stainless-steel current collectors, with an LFP ss 16 mm and TiS_2 ss 12 mm, both mounted on a metal mesh holder to ensure mechanical support and acoustic coupling. Electrolyte for the reference experiment is an aqueous 2 m (molal) LiTFSI solution ("2 m LiTFSI in H_2O"). The electrochemical protocol is cyclic voltammetry (CV) over a 0.2–1.4 V window at a 0.15 mV s^{-1} scan rate, with 1.4 V denoted "1,4 V" in European notation.

2.1.2 AE acquisition and event definition

Acoustic emission was recorded on channel CH1 (the first AE input). Each AE "event" is a triggered waveform segment containing 3000 samples spanning a fixed pre-trigger and post-trigger window around the detection time. This fixed-length capture preserves the onset and immediate relaxation dynamics while standardizing inputs for learning-based models.

2.1.3 Datasets

We analyze two complete experiments that share the same cell configuration and AE setup. One experiment uses a 2 m LiTFSI electrolyte; the other uses a 5 m electrolyte. Each experiment contains on the order of AE events ($\approx$ 40k per run). Every event is a fixed-length waveform with 3000 samples. We adopt a cross-experiment split: the 2 m run is used for training, and the 5 m run is used for validation and out-of-distribution checks.

2.2 Preprocessing

2.2.1 Inventory and time indexing (common to all analyses)

We first build an event inventory by scanning raw AE directories and recording, for each event CSV, its absolute file path, experiment identifier, and date; the result is stored as a Parquet table. Filenames contain a trailing integer token $\langle idx \rangle$ (e.g., `rawAE_..._<idx>.csv`), which we interpret at a nominal sampling rate $f_s = 10$ MHz to obtain a per-event start time via $t_{\text{start}} = idx/f_s$. This timestamp is later used for alignment with the electrochemical log.

2.2.2 Electrochemical alignment and coarse phase labels (common)

The CV log is tab-delimited and contains time (hours), voltage, current, and cycle index. Time is converted to microseconds. Each AE event is matched to the nearest CV timestamp using its start time. The coarse electrochemical state is defined by the sign of the current: charging ($I > 0$), discharging ($I < 0$), or OCV ($|I| \approx 0$). We then create a label of the form `<cycle>_<state>` for downstream correlation.

2.2.3 Feature-space preprocessing (for classical baselines and diagnostics)

When computing classical AE descriptors and CV-aligned summaries, we work in the original amplitude domain. Unit switches between V and mV are for convenience only. For each event we read time t and amplitude amp (V). We then compute: RMS (mV); energy as $\sum amp^2/fs$ ($V^2 \cdot s$); energy MARSEenergy $\times 10^6$ ($\mu s \cdot V^2$; stored as `energy_marse`); peak amplitude and a μV-referenced dB value; duration using a dynamic threshold at 5-σ; rise time from the first zero crossing to the peak; activity-rate ($|amp| > 0$ mV per unit time); and FFT-based peak and centroid frequencies (kHz). We apply a light outlier filter (`energy_marse < 100`) before tabulation. The resulting table is saved as `events_features.parquet` for exploratory plots and sanity checks.

2.2.4 Representation-learning preprocessing (VQ-VAE inputs and model-derived features)

For representation learning we feed raw waveforms to a trained 1-D VQ-VAE encoder–quantizer–decoder. Each event is standardized independently by per-event z-scoring,

$$x_{\text{norm}} = \frac{x - \mu}{\sigma + 10^{-6}}, \tag{1}$$

where (μ, σ) are the event's mean and standard deviation retained for reporting errors back in the original amplitude domain. Per-event z-score stabilizes training by controlling amplitude scaling while preserving within-event shape, and allows amplitude-domain metrics post hoc via stored (μ, σ). We avoid global normalization (e.g., per-experiment mean/variance) to reduce sensitivity to sensor coupling changes and to prevent information leakage across the train/-validation split. No whitening or cross-event normalization is used, avoiding leakage across experiments.

2.2.5 Data Exploration

We performed light exploratory checks on the extracted AE features to validate threshold settings and to understand coarse time–frequency trends before representation learning. With the dynamic threshold applied at the waveform level, derived quantities such as `duration_us`, `rise_time_us`, `counts`, and `RA_value` change as expected. Time–frequency scatter plots

against cycling time (Fig. 1) show compact frequency bands during early OCV and more dispersed bands during charge/discharge, with occasional > 700 kHz spikes (candidate high-energy micro-events). Detailed figures and additional comments are provided in the Appendix.

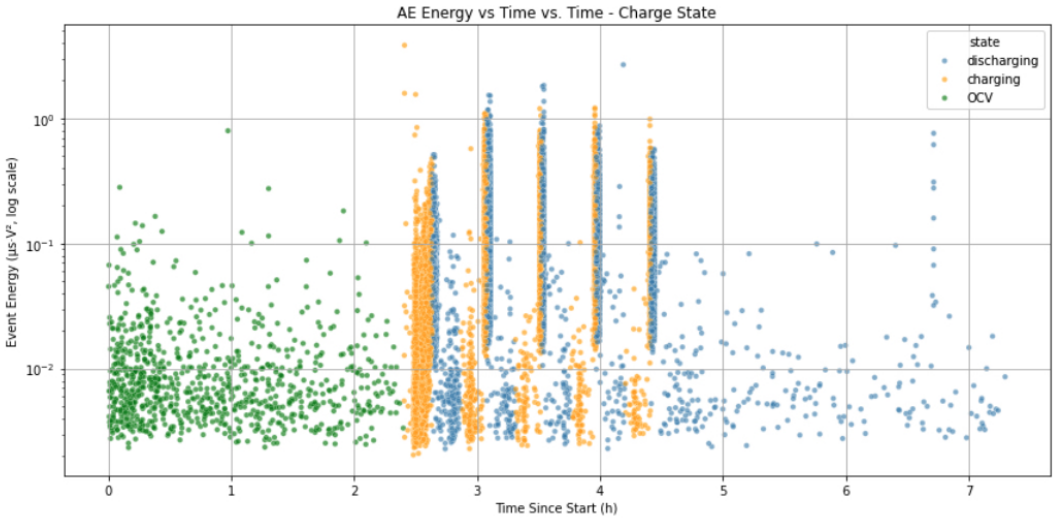


Figure 1: Time-frequency behavior across OCV/charge/discharge states

2.3 Model

2.3.1 Architecture (1-D VQ-VAE, EMA codebook)

We use a 1-D convolutional VQ-VAE with an EMA-updated codebook. See Appendix Fig. 10 for a schematic. Inputs are per-event z-scored AE waveforms of length $L = 3000$ (one channel). The encoder comprises three Conv1d blocks with kernel size 4, stride 2, padding 1 and ReLU activations, mapping

$$(B, 1, 3000) \xrightarrow{64} (B, 64, 1500) \xrightarrow{128} (B, 128, 750) \xrightarrow{128} (B, 128, 375),$$

so the latent length is $L' \approx 375$ and the embedding dimension is $D = 128$. The tensor is transposed to (B, L', D) and quantized by a vector-quantizer with $K = 64$ codewords using EMA updates and a straight-through estimator for gradients. The decoder mirrors the encoder with three ConvTranspose1d blocks (kernel 4, stride 2, padding 1, ReLU except the last layer), producing $\hat{x}_{\text{norm}} \in \mathbb{R}^{B\times 1\times 3000}$. We track code indices and usage for each event and compute dataset-level histograms for analysis.

2.3.2 Objective

The training loss is the sum of a reconstruction term in the normalized domain and a VQ commitment term,

$$\mathcal{L} = \frac{1}{BL} \sum \|\hat{x}_{\text{norm}} - x_{\text{norm}}\|_2^2 + \beta \|\text{sg}[z_e] - z_q\|_2^2, \tag{2}$$

with $\text{sg}[\cdot]$ the stop-gradient operator, z_e the encoder outputs, and z_q the quantized embeddings. A small entropy regularizer on the usage histogram was optionally used to discourage codebook collapse.

2.3.3 Data split and training protocol

Training data: `CH1_LFP-TiS2_2m_2025-02-03` (40,880 events; 3000 samples/event).
Validation data: `CH1_LFP-TiS2_5m_2025-01-28` (7,565 events; 3000 samples/even).

Optimization uses Adam with a small learning-rate grid; mini-batches are shuffled across cycles. We train on the 2 m dataset and evaluate out-of-distribution on the 5 m dataset to probe generalization across operating conditions.

2.3.4 Model-derived representations

Beyond reconstructions, we export two representations per event for downstream analysis: (i) bag-of-codes histograms $h \in \mathbb{R}^K$ (counts over the L' latent positions, normalized to sum to 1), and (ii) codebook barycenters $z = h^T E \in \mathbb{R}^D$, where E stacks the K embedding vectors. These are used for visualization and clustering.

2.4 Metrics

2.4.1 Reconstruction error (normalized and amplitude domains)

- MSE (normalized):

$$\text{MSE} = \frac{1}{BL} \sum_{b,\ell} \left(\hat{x}^{(b)}_{\text{norm}}[\ell] - x^{(b)}_{\text{norm}}[\ell]\right)^2. \tag{3}$$

- NMSE (original amplitude): for each event, recover $\hat{x} = \mu + \sigma \hat{x}_{\text{norm}}$ and $x = \mu + \sigma x_{\text{norm}}$, then

$$\text{NMSE} = \frac{\frac{1}{L}\sum_\ell (\hat{x}[\ell] - x[\ell])^2}{\frac{1}{L}\sum_\ell x[\ell]^2}. \tag{4}$$

We report medians over events (with IQR when applicable) to reduce sensitivity to outliers.

2.4.2 Codebook usage diversity

Let p_k be the normalized usage of code k aggregated over a dataset. We measure diversity via perplexity,

$$\text{Perplexity} = \exp\Big(-\sum_{k=1}^{K} p_k \log p_k\Big), \tag{5}$$

which equals K for uniform use and drops as usage concentrates.

3 Results

We report reconstruction metrics and codebook statistics, then study clustering in the learned representation. For completeness, we include feature-space PCA baselines as a point of comparison.

3.1 Reconstruction quality

Observed values. Train: MSE = 8.13×10^{-7}, NMSE = 0.0034.
Validation: MSE = 4.09×10^{-7}, NMSE = 0.0206.

Lower MSE on validation but higher NMSE indicates the 5 m experiment has lower absolute amplitudes (making normalized errors look small) and some shape/energy shift that raises relative error—consistent with cross-experiment domain shift.

In addition to MSE/NMSE, we show qualitative reconstructions for randomly sampled events from the train and validation sets (Fig. 2). For each event we plot the original waveform, the reconstruction, the residual $x - \hat{x}$, and a 100–150-sample zoom-in window. All panels are in the original amplitude domain with a shared y-axis per row to allow visual comparison.

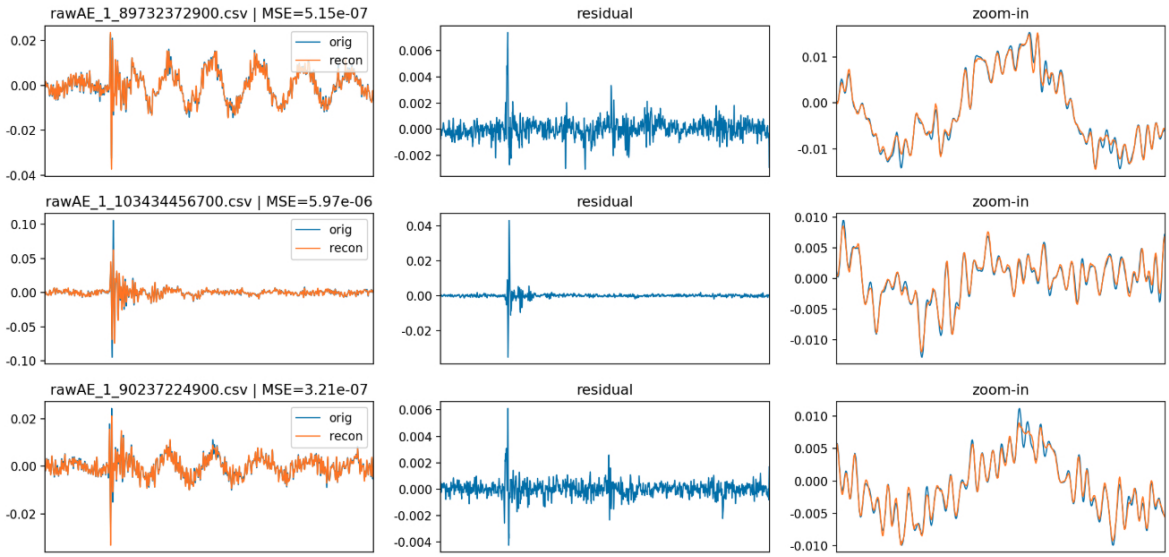


Figure 2: Qualitative reconstructions. Rows: randomly sampled events (seed=42). Columns: (a) original vs. reconstructed waveform; (b) residual $x - \hat{x}$; (c) zoom-in around the largest local peak. Titles report per-event MSE (normalized domain). This view complements MSE/NMSE by revealing where errors concentrate (e.g., sharp transients vs. slowly varying envelopes) and highlights differences between train and validation distributions.

3.2 Codebook usage

With $K = 64$, we observe:

- Train perplexity ≈ 40.9
- Validation perplexity ≈ 51.7

The higher perplexity on the 5 m dataset suggests richer or more varied code activation patterns under that condition. We visualize (i) **codebook usage** per dataset and (ii) **representative reconstructions** for a few codes.

Figure 3 compares usage histograms for the 2 m (left) and 5 m (right) experiments. Bars are normalized to probabilities ($\sum_k p_k = 1$); we annotate dataset-level perplexity and mark the top-5 most frequent codes.

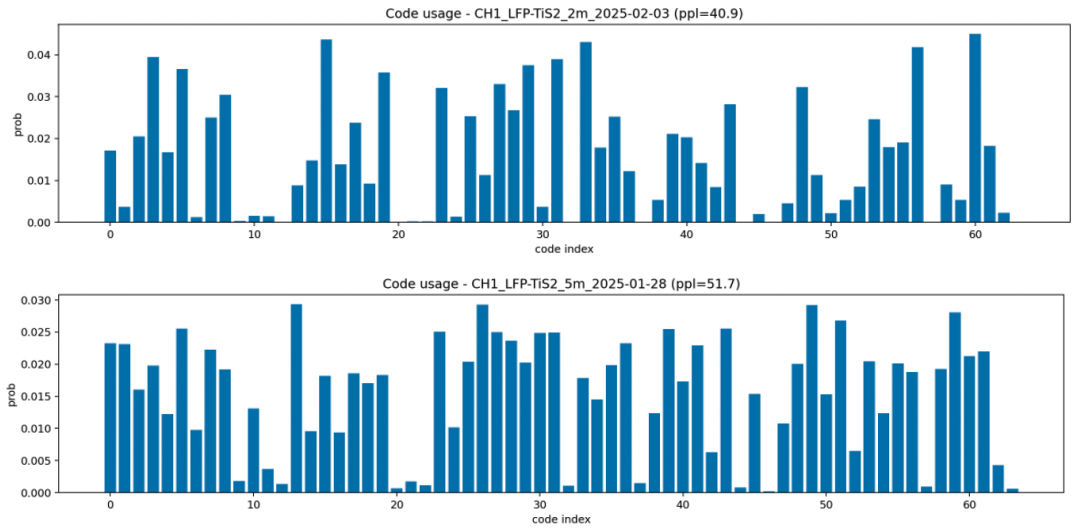


Figure 3: Codebook usage comparison across experiments. Left: train dataset; Right: validation dataset. Bars show normalized code usage p_k (counts over latent positions divided by L' and aggregated over all events). Perplexity is reported in the title; the top-5 codes by usage are highlighted. This reveals a broader spread at 5 m (higher perplexity), indicating more diverse code activations under that condition.

The model learns a compact representation that reconstructs training events with low relative error and exhibits condition-dependent code usage. The divergence between normalized-domain MSE and amplitude-domain NMSE across experiments underscores the need to report **both** metrics when amplitudes vary across operating conditions. Figure 4 shows reconstructions for selected codes: for each code, we display the average waveform of events where the code is highly expressed together with several single-event reconstructions and their residuals. These plots support qualitative inspection of learned motifs and condition-dependent differences.

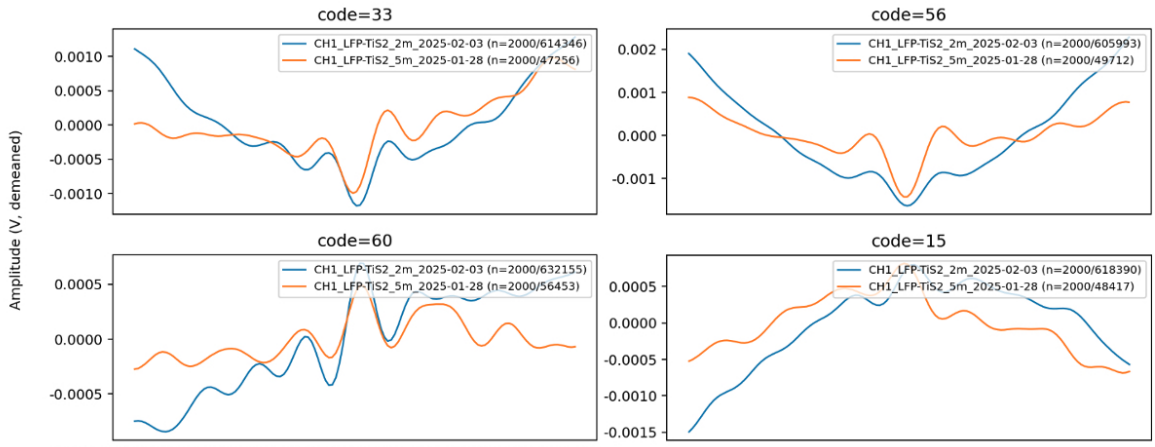


Figure 4: Reconstructions for selected codes. For codes $\{15, 33, 56, 60\}$ (or dataset-specific frequent codes), we plot: (A) the mean waveform of events where the code is highly expressed, (B) a few randomly sampled single-event pairs $(x, \hat{x})$, and (C) residuals $x - \hat{x}$. All waveforms share the same y-axis scale and are shown in the original amplitude domain (restored from x_{norm}). These examples illustrate the kinds of local motifs each code captures and where reconstruction errors concentrate.

3.3 Clustering

Representations For each event, the VQ-VAE produces a length-L' sequence of code indices. We form a *bag-of-codes* histogram $h \in \mathbb{R}^K$ by counting occurrences and normalizing to sum to 1 ($h_k = \text{count}(k)/L'$). Unless stated, $K = 64$ and $L' \approx 375$. We also compute a *codebook barycenter* $z = h^\top E \in \mathbb{R}^{128}$, where $E \in \mathbb{R}^{K\times 128}$ stacks the learned embeddings. Both are exported per event for downstream analysis.

Dimensionality reduction and clustering For each event we convert the discrete code sequence to a 64-D bag-of-codes (BoC) probability vector and embed these vectors with UMAP (cosine; $n_{\text{neighbors}} = 30$; $min_dist = 0.1$) [5]. In both experiments the embedding forms a leaf-like, near-continuous manifold, suggesting smooth variation along a few latent axes (energy/burstiness/spectral balance) rather than separable classes.

In this representation space, HDBSCAN shows many small to medium clusters that tile the broad lobe and flags a modest number of outliers (Fig. 5); the clusters follow local density variations rather than distinct modes [6, 7].

In contrast, K-Means (k=5) slices the manifold into elongated wedges aligned with the main gradient (Fig. 6), producing partitions that are sensitive to initialization and exhibit low separability.

Together, these outcomes indicate that hard partitioning is not the data's natural organization; segmenting along the main axis or using continuous code-usage summaries is more faithful than enforcing discrete blobs.

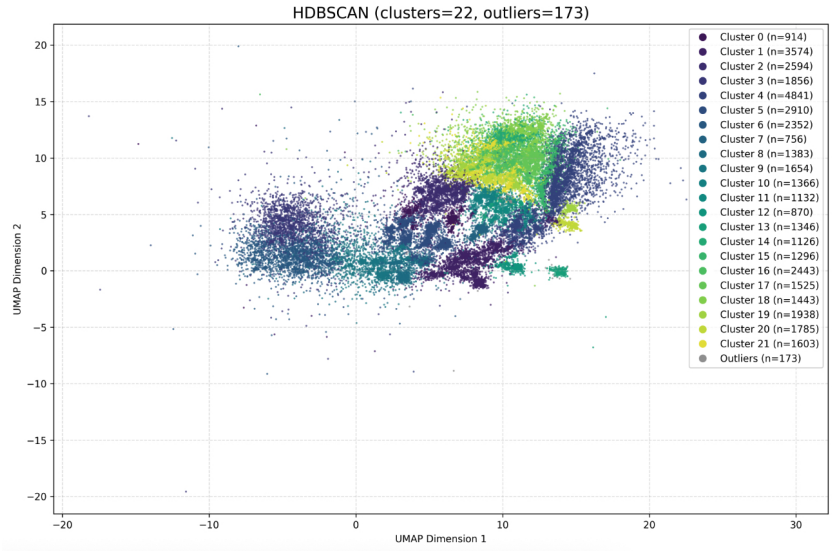


Figure 5: HDBSCAN on UMAP. BoC UMAP colored by HDBSCAN labels: many small/medium clusters plus a few outliers, reflecting smooth density variation rather than well-separated modes.

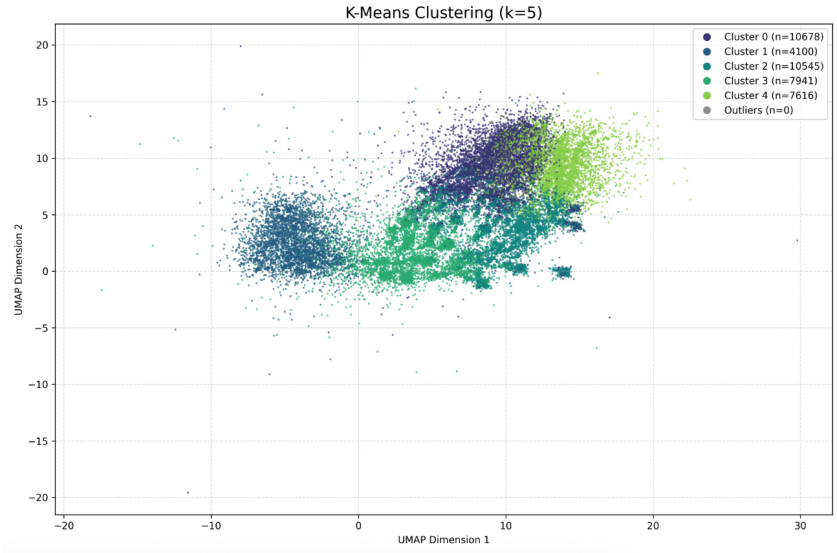


Figure 6: K-Means ($k = 5$) on UMAP. BoC UMAP colored by K-Means: the leaf-shaped manifold is cut into elongated wedges aligned with the global gradient, indicating weak cluster separability.

Segment–reaction alignment We first summarize the UMAP geometry by a one-dimensional main-axis coordinate s_{coord}: apply PCA to the 2-D UMAP embedding and take the first principal direction. Coloring points by s_{coord} reveals a smooth gradient from the thin tail to the broad lobe, i.e., a monotonic progression along the manifold (Fig. 7). We then partition s_{coord} into six equal-frequency segments ($k = 6$), which provides a simple, length-preserving way to discretize this continuous axis.

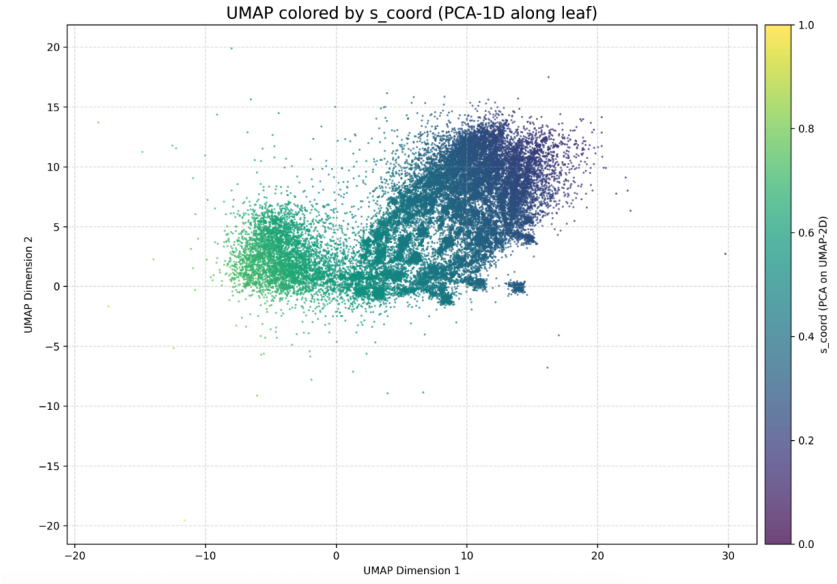


Figure 7: UMAP colored by s_{coord}. Main-axis coordinate obtained by PCA on the 2-D UMAP. The smooth color gradient from tail to lobe shows a monotonic progression along the manifold.

To assess whether the acoustic axis relates to electrochemical behavior, we assign weak reac-

tion labels using quantile-based rules that combine AE energy, spectral centroid, duration/rise time, and voltage/current cues. We obtain the reaction–by–segment conditional distribution $P(\text{reaction} \mid \text{segment})$.

The stacked bars (Fig. 8) show a monotonic shift across segments: early segments (low s_{coord}) are enriched in transient, high-energy bursts, middle segments contain mixed/transition behavior, and late segments (high s_{coord}) are dominated by steady/OCV or main charging activity. This pattern indicates that the learned acoustic main axis is physically meaningful and tracks electrochemical progression. Practically, it also explains why hard clustering is brittle: events are organized along a continuum, so segmenting along s_{coord} is a more faithful summary than enforcing discrete blobs.

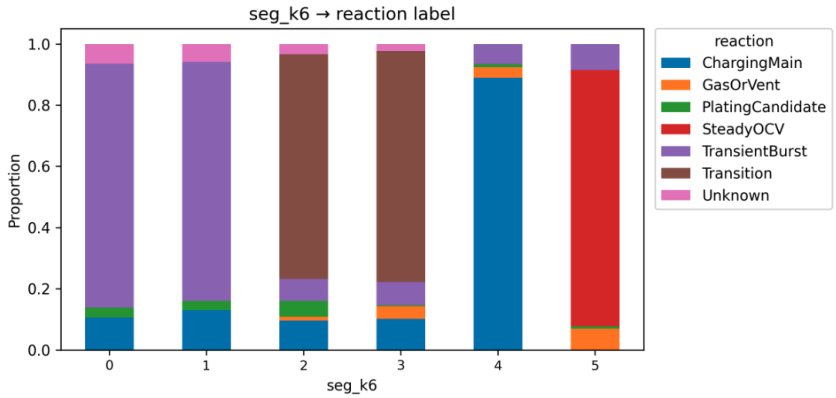


Figure 8: Reaction–by–segment. Proportion of weak reaction labels within each s_{coord} segment. Distributions shift from transient, high-energy bursts (left) to steady/OCV regimes (right), supporting alignment between the acoustic axis and electrochemical behavior.

3.4 Feature-space baselines (PCA)

As a baseline in the handcrafted-feature space, we standardized six descriptors (duration, energy/energy MARSE, rise time, RMS, peak frequency, centroid frequency) and ran PCA for dimensionality reduction followed by LDA using three amplitude bands (low/medium/high) for supervision.

PCA indicates that **PC1** overall amplitude/energy axis (high loadings on energy MARSE and RMS; negative on spectral centroid), **PC2** dominated by peak frequency (with some RMS/-MARSE), and **PC3** combines peak frequency with rise time. Later components capture narrower contrasts (e.g., spectral centroid in PC5). The cumulative variance explained plateaus after 4–5 PCs, suggesting a compact feature manifold (Fig. 9). These baselines recover the coarse structure but also motivate learned representations to handle overlapping sources and distribution shift more robustly.

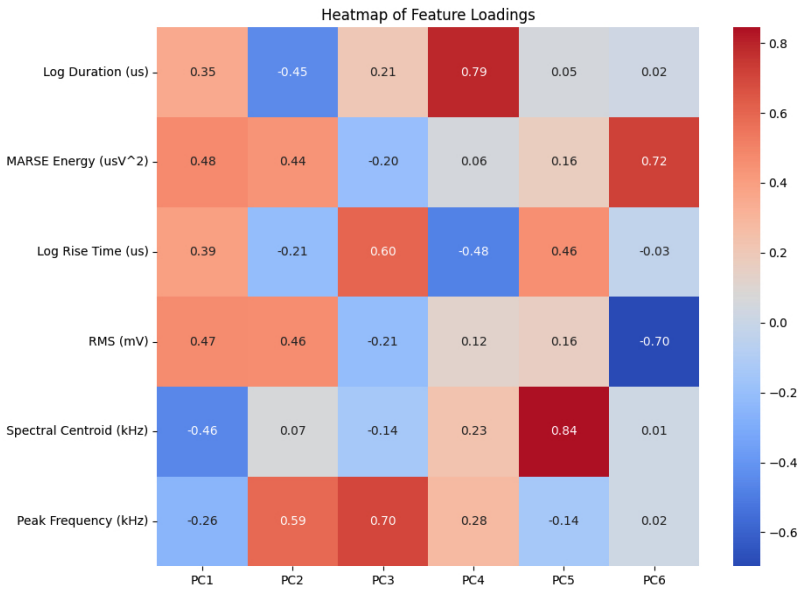


Figure 9: PCA projection (first three PCs) with amplitude bands.

4 Next Steps

Future improvements to clustering could reduce phase sensitivity and background bias. First, we can operate in phase-invariant transformed spaces using magnitude Fourier spectra or time-frequency maps (STFT/CWT). This shifts comparisons toward spectral composition rather than timing. Such representations are standard for non-stationary signals and are well reviewed in the wavelet literature [8].

Second, we can subtract an experiment-specific background. Estimate a template from non-event segments and remove it from event features (spectral or time-domain). This focuses clustering on deviations from steady acoustics. The idea is analogous to spectral subtraction in noise suppression [9].

Together, these adjustments are expected to give more stable groupings and clearer links to electrochemical behavior.

5 Conclusion

We built an end-to-end pipeline that learns discrete, label-free representations of battery AE waveforms with a 1-D VQ-VAE and summarizes events using bag-of-codes statistics. Reconstructions are stable. Code usage shifts across conditions (perplexity is higher at 5 m). The learned main axis also tracks electrochemical progression, which explains why classical hard clustering fragments the data while segment-wise analyses yield coherent trends. Overall,

these results show that unsupervised representation learning is a practical basis for AE analytics and motivate phase-robust, transformed-space clustering and residual-focused features to link acoustic motifs to underlying electrochemical mechanisms.

6 Acknowledgments

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Appendix

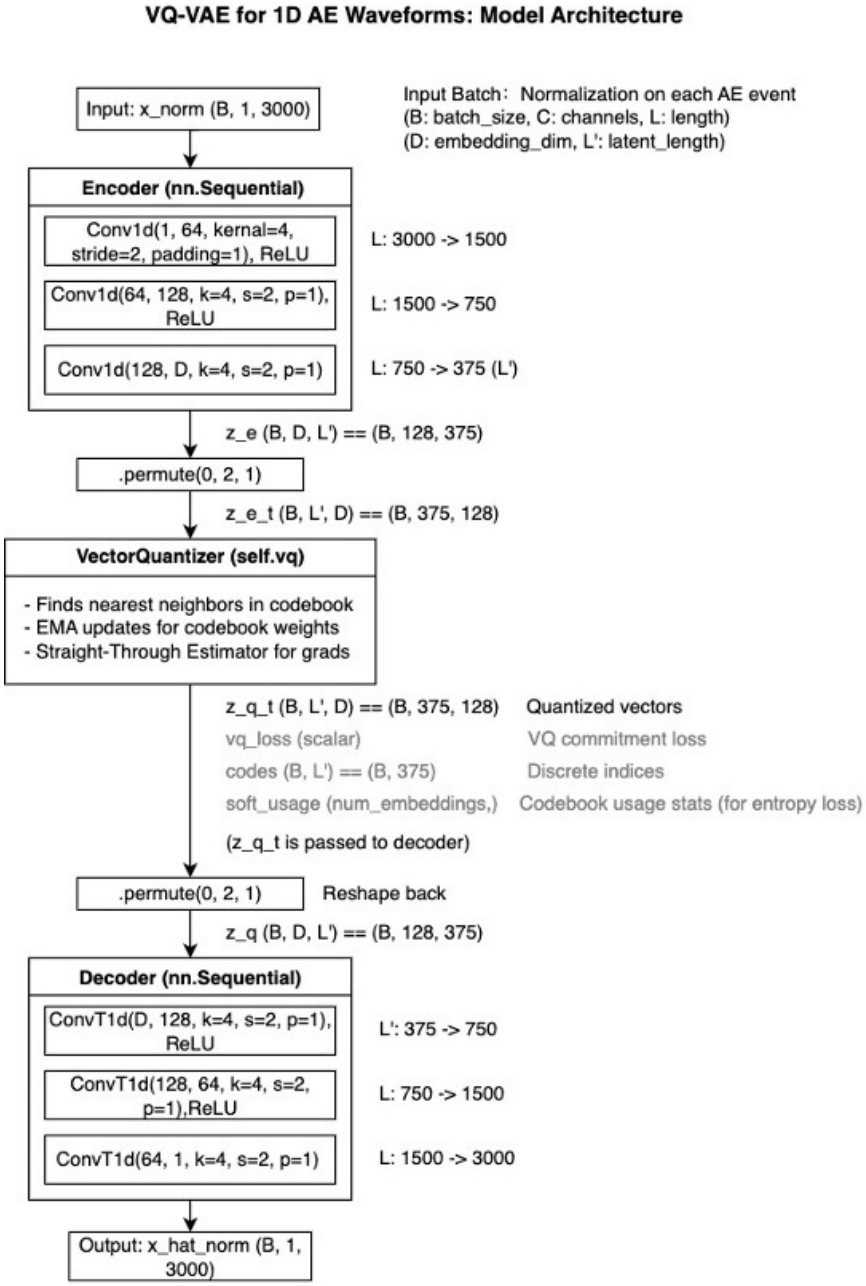


Figure 10: 1-D VQ-VAE architecture. Encoder (stride-2 $\times$ 3) maps 3000 samples to a latent length of ≈ 375; a $K = 64$ EMA codebook quantizes $D = 128$ vectors; the decoder mirrors the encoder to reconstruct the waveform.